
EEG-to-Image Retrieval via Asymmetric Brain–Image Co-Adaptation

Chi Kit Wong

Sida Zhang

Zhiyi Chen

Ziyi Guo

AI Thrust

HKUST (GZ)

{ckwong627, szhang644, zchen986, zguo315}@connect.hkust-gz.edu.cn

Project repository: [Chikit-WONG/Human-Centered_AI_Group_Project](#)

Abstract

Electroencephalography (EEG) offers a non-invasive view of millisecond-scale neural responses, but exact-image retrieval remains difficult because EEG is noisy, subject-dependent, and semantically misaligned with pretrained image features. We study 200-way zero-shot EEG-to-image retrieval on THINGS-EEG2 and propose an asymmetric co-adaptation strategy: a lightweight EEG encoder is trained rapidly while low-rank adapters update a pretrained CLIP image encoder at a slower rate. Across ten subjects and five fixed seeds, our method obtains $86.66 \pm 0.69\%$ Top-1 and $98.38 \pm 0.14\%$ Top-5 accuracy. A strictly paired subject-08 ablation improves Top-1 from 29.5% with a frozen image encoder to 61.5% with equal-rate adaptation and 89.0% with asymmetric adaptation, isolating the contribution of co-adaptation from the EEG architecture. A focused hyperparameter study raises the seed-42 ten-subject Top-1 mean from 85.85% to 87.70%, while strict leave-one-subject-out evaluation reaches 20.40% Top-1 and exposes a substantial subject-shift gap. Finally, controlled matching experiments distinguish standard independent retrieval from batch-level assignment. Hungarian and Sinkhorn decoding can improve a known one-to-one session, but their advantage weakens when images are missing or multiple real EEG measurements correspond to one image. These results support asymmetric representation learning while clarifying the assumptions and limitations of session-aware post-processing.

1 Introduction

Human visual experience produces rapid neural activity that can be measured non-invasively with EEG. The resulting signals have high temporal resolution and are comparatively accessible for human–computer interaction, but their low signal-to-noise ratio, limited paired data, and strong inter-person variability make visual decoding difficult. THINGS-EEG2 provides repeated EEG responses to natural object images and has enabled controlled evaluation of neural representations at scale [1].

We focus on *exact EEG-to-image retrieval*: given the EEG response evoked by an unseen stimulus, rank the correct image among 200 candidates. This differs from image generation and from category-level recognition. The output is a ranked gallery, and the primary metrics are independent-query Top-1 and Top-5 accuracy. A central challenge is the representation mismatch between a small, noisy EEG modality and an image encoder pretrained on large visual corpora. Training only

an EEG encoder forces neural signals into a fixed machine-vision space; fully fine-tuning the image encoder risks destroying useful pretrained structure.

Our method addresses this tension with asymmetric co-adaptation. The EEG encoder and image-side LoRA adapters [2] share the same contrastive objective, but the EEG branch receives a larger learning rate. This design is motivated by two-time-scale optimization [3]: the randomly initialized EEG branch must learn rapidly, whereas the pretrained image branch should move conservatively. Figure 1 shows both training and retrieval. We make four contributions:

- an efficient dual-side alignment framework that combines a compact EEG encoder with conservative CLIP image-side adaptation;
- a strictly paired A/B/C ablation that holds the EEG architecture, initialization, data, seed, and training budget fixed while changing only image-side optimization;
- subject-dependent five-seed evaluation, a focused rank–learning-rate study, and strict leave-one-subject-out (LOSO) evaluation;
- a fairness-controlled comparison of five decoders on identical score matrices, including robustness tests that deliberately violate one-to-one correspondence.

The distinction between representation quality and assignment post-processing is important. Independent retrieval asks which image best matches an EEG query. In contrast, Hungarian, stable, and transport-based decoders use information about the whole evaluation batch. We therefore report these quantities separately and do not relabel assignment accuracy as standard retrieval Top-1 or Top-5.

2 Related Work

EEG–image alignment. NICE demonstrated that contrastive alignment can learn image-related EEG representations for zero-shot object recognition [4]. ATM-S introduced temporal and spatial attention and used EEG embeddings for retrieval and guided image reconstruction [5]. UBP reduces the vision–brain information gap by adapting image content with an uncertainty-aware blur prior [6]. NeuroBridge uses cognitive-prior augmentations and a shared semantic projector for bidirectional alignment [7]. Shallow Alignment argues that intermediate image features better match the granularity of neural signals than final semantic features [8]. These approaches motivate a broader question: should the image target remain fixed, or should it adapt jointly with the brain representation? Our work tests the latter under a deliberately conservative update rule.

Pretrained image representations and parameter-efficient adaptation. CLIP learns transferable image–text representations from large-scale supervision [9]. We use its ViT-B/32 image encoder as the visual prior. LoRA freezes the original weight matrix and optimizes a low-rank residual, reducing trainable parameters and limiting drift [2]. Our use of LoRA is modality-specific: the adapters reshape the image target only enough to improve EEG alignment, while the EEG branch remains the primary learner.

Global assignment and optimal transport. The Hungarian algorithm solves a linear one-to-one assignment problem [10]; Gale–Shapley stable matching produces a stable pairing from ranked preferences [11]; and Sinkhorn iterations approximate entropy-regularized optimal transport by matrix scaling [12, 13]. Such decoders can exploit known batch structure, but they change the inference problem. We evaluate them as optional offline analyses, not as replacements for independent retrieval.

3 Task and Method

3.1 Retrieval task

Let $x_i \in \mathbb{R}^{17 \times 250}$ denote the preprocessed EEG response for query i , and let v_j denote candidate image j . The EEG encoder f_θ and image encoder g_ϕ produce L2-normalized embeddings

$$z_i = \frac{f_\theta(x_i)}{\|f_\theta(x_i)\|_2}, \quad y_j = \frac{g_\phi(v_j)}{\|g_\phi(v_j)\|_2}. \quad (1)$$

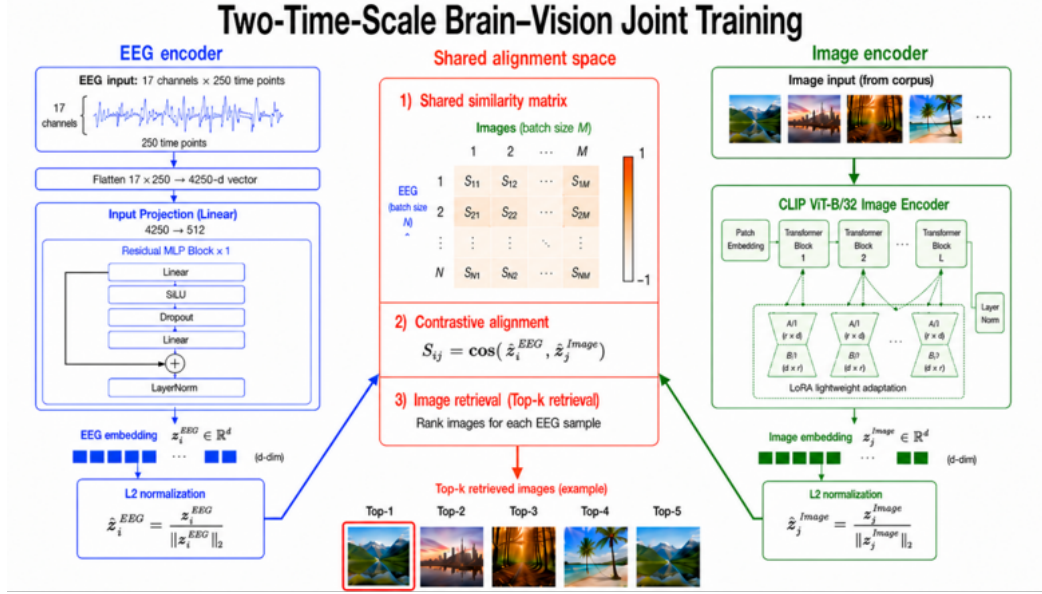


Figure 1: Detailed architecture of our method. During training, a compact EEG encoder and image-side LoRA adapters map paired EEG trials and stimulus images into a shared CLIP-aligned space and are optimized with a contrastive loss. The EEG branch learns at a faster rate than the image branch. During inference, a query EEG embedding is compared with a precomputed image bank to produce an ordinary Top-1/Top-5 ranking.

Their scaled cosine score is $S_{ij} = \exp(\tau) z_i^\top y_j$, where τ is learnable. Standard independent retrieval ranks every row of S separately. For ground-truth image index t_i , Top- k is

$$\text{Top-}k = \frac{1}{Q} \sum_{i=1}^Q \mathbb{I}[t_i \in \text{TopK}_k(S_{i,:})]. \quad (2)$$

No knowledge that gallery images are unique is used in Equation 2.

3.2 EEG encoder

The input uses 17 posterior channels and 250 time samples. We flatten the 17×250 response, project it to 512 dimensions, and apply one residual two-layer MLP block with SiLU activation and layer normalization. This intentionally compact architecture limits capacity relative to the image backbone. The output is L2-normalized as in Equation 1. Because the strict A/B/C experiment in Section 5.2 uses this identical EEG architecture in all conditions, its differences cannot be attributed to replacing the EEG encoder.

3.3 Conservative image-side LoRA adaptation

The image branch starts from a pretrained CLIP ViT-B/32 image checkpoint. Original CLIP weights remain frozen, and LoRA adapters are inserted into linear modules. The reference setting uses rank $r = 32$, scaling $\alpha = 32$, zero dropout, and Gaussian adapter initialization. If W_0 is a frozen image-layer matrix, the adapted layer is

$$h' = W_0 h + \frac{\alpha}{r} B A h, \quad (3)$$

where A and B are trainable low-rank factors. Unlike frozen-target alignment, Equation 3 allows the image space to move toward EEG-relevant structure while retaining the pretrained backbone.

3.4 Contrastive objective and asymmetric optimization

For a minibatch of N paired EEG–image examples, we minimize symmetric cross-entropy over the similarity matrix:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N \left[-\log \frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} - \log \frac{e^{S_{ii}}}{\sum_j e^{S_{ji}}} \right]. \quad (4)$$

The reference EEG learning rate is 5×10^{-4} , whereas the image-adapter learning rate is 5×10^{-5} , a ratio of 10:1. Both branches receive gradients from Equation 4, but their update magnitudes differ. This is the operational meaning of asynchronous or two-time-scale learning in our method; it is not alternating training and does not use a separate objective.

3.5 Optional batch-level matching

Given a complete score matrix, independent retrieval uses $\hat{t}_i = \arg \max_j S_{ij}$. Greedy one-to-one decoding processes high-confidence queries first and removes selected images. Hungarian decoding instead solves

$$\hat{\pi} = \arg \max_{\pi \in \text{Perm}(N)} \sum_{i=1}^N S_{i, \pi(i)}, \quad (5)$$

for a square session, while rectangular matrices leave surplus rows or columns unmatched. Stable matching uses query-proposing Gale–Shapley preferences derived from S . Sinkhorn applies log-domain scaling to an entropy-regularized transport plan, followed by row-wise plan decoding. We use temperature 0.05, at most 500 iterations, and a strict residual tolerance of 10^{-8} .

These structured methods naturally yield one assignment per query rather than a ranked list. We consequently report their *assignment accuracy*, matched/unmatched counts, and convergence diagnostics. Top-5 is reported only for independent retrieval.

4 Experimental Protocol

4.1 Dataset, splits, and metrics

We use THINGS-EEG2 [1], containing EEG responses to natural object images from ten subjects. Following the standard zero-shot split, training and test concepts are disjoint, and evaluation uses 200 unseen test images. We select 17 posterior channels. Training inputs average four trials; standard test inputs average 80 repeated trials. Subject-dependent experiments train one model per subject. Strict LOSO trains on nine subjects and evaluates the tenth, producing ten folds. Repeated-trial averaging improves signal-to-noise ratio and is explicitly distinguished from single-trial decoding.

The primary subject-dependent result averages seeds $\{42, 43, 44, 45, 46\}$. We report the mean and sample standard deviation of the five seed-level ten-subject macro means. LOSO and the focused formal confirmation use seed 42 and are labeled single-seed. Random Top-1 and Top-5 levels in a 200-image gallery are 0.5% and 2.5%, respectively.

4.2 Optimization and checkpoint rules

The reference model uses AdamW, cosine learning-rate decay, weight decay 0.05, batch size 512, evaluation batch size 100, 25 epochs, bfloat16, gradient checkpointing, and a gradient-norm limit of 1.0 applied after back-propagation. The main historical grid uses the fixed final epoch to avoid selecting on test data. For fairness-suite NICE and ATM-S re-evaluations, one training run is performed and the checkpoint is selected by validation loss. Our method’s fairness matrix reuses its fixed final checkpoint. We report these differences because the suite is a controlled implementation comparison, not a claim of perfect reproduction.

4.3 Non-bijective robustness scenarios

Matching assumptions are tested on identical scores under: the standard 200×200 session; removal of 5 or 10 EEG queries; removal of 5 or 10 images; asymmetric removal of both sides; duplication

Table 1: Per-subject 200-way EEG-to-image retrieval accuracy (%). Intra trains and tests within one subject; Inter uses leave-one-subject-out evaluation. Published comparison rows are contextual. Our intra-subject values average five seeds; our inter-subject values are strict seed-42 LOSO ([†]).

Method	Metric	Sub1	Sub2	Sub3	Sub4	Sub5	Sub6	Sub7	Sub8	Sub9	Sub10	Avg.
<i>Intra-subject: train and test on the same subject</i>												
NICE [4]	Top-1	13.2	13.5	14.5	20.6	10.1	16.5	17.0	22.9	15.4	17.4	16.1
	Top-5	39.5	40.3	42.7	52.7	31.5	44.0	42.1	56.1	41.6	45.8	43.6
ATM-S [5]	Top-1	25.6	22.0	25.0	31.4	12.9	21.3	30.5	38.8	34.4	29.1	27.1
	Top-5	60.4	54.5	62.4	60.9	43.0	51.1	61.5	72.0	51.5	63.5	58.1
UBP [6]	Top-1	41.2	51.2	51.2	51.1	42.2	57.5	49.0	58.6	45.1	61.5	50.9
	Top-5	70.5	80.9	82.0	76.9	72.8	83.5	79.9	85.8	76.2	88.2	79.7
NeuroBridge [7]	Top-1	50.0	63.2	61.6	61.4	54.8	69.7	62.7	71.2	64.0	73.6	63.2
	Top-5	77.6	90.6	91.1	90.0	85.0	92.9	88.8	95.1	91.0	97.1	89.9
Shallow [8]	Top-1	75.4	87.3	82.9	79.1	74.9	90.2	79.0	86.9	81.3	89.3	82.6
	Top-5	94.3	99.0	98.3	96.5	96.4	99.2	97.3	99.4	97.8	99.2	97.7
Our method	Top-1	84.0	89.7	85.2	83.4	84.0	92.8	84.8	90.9	80.3	91.5	86.66
	Top-5	96.6	99.7	97.4	96.5	98.3	99.8	98.0	99.9	97.7	99.9	98.38
<i>Inter-subject: leave one subject out for test</i>												
NICE [4]	Top-1	7.6	5.9	6.0	6.3	4.4	5.6	5.6	6.3	5.7	8.4	6.2
	Top-5	22.8	20.5	22.3	20.7	18.3	22.2	19.7	22.0	17.6	28.3	21.4
ATM-S [5]	Top-1	10.5	7.1	11.9	14.7	7.0	11.1	16.1	15.0	4.9	20.5	11.9
	Top-5	26.8	24.8	33.8	39.4	23.9	35.8	43.5	40.3	22.7	46.5	33.8
UBP [6]	Top-1	11.5	15.5	9.8	13.0	8.8	11.7	10.2	12.2	15.5	16.0	12.4
	Top-5	29.7	40.0	27.0	32.3	33.8	31.0	23.8	32.2	40.5	43.5	33.4
NeuroBridge [7]	Top-1	23.2	21.2	13.2	17.0	14.5	25.0	15.3	20.1	13.7	27.2	19.0
	Top-5	52.4	49.3	36.5	45.3	37.7	55.0	45.1	44.9	36.5	56.3	45.9
Shallow [8]	Top-1	24.6	31.3	11.4	19.9	19.0	24.1	18.6	17.6	23.3	34.6	22.4
	Top-5	54.7	61.5	31.1	48.8	45.5	49.8	51.6	46.7	54.9	63.2	50.8
Our method[†]	Top-1	27.0	31.5	13.0	20.5	14.0	15.5	17.0	15.0	20.0	30.5	20.40
	Top-5	59.0	63.5	34.0	50.5	37.5	47.5	43.5	42.5	50.5	64.5	49.30

of 10 or 20 gallery images; addition of unrelated distractor images; addition of EEG queries whose correct image is absent; and combinations of missing, duplicated, and distractor items.

For a realistic duplicate-EEG condition, we do not copy score rows. Within every recording session, the first ten trials for an image are averaged into EEG-A-1 and the remaining ten into EEG-A-2. The two averages are disjoint real measurements. The base uses 200 EEG-A-1 queries; adding EEG-A-2 for 10 or 20 selected images produces 210×200 and 220×200 matrices. Their absolute scores are compared only within this 20-trial-average suite, not against the standard 80-trial test.

5 Results

5.1 Subject-dependent and cross-subject retrieval

Table 1 follows the requested per-subject format. Literature rows use reported values under closely related 200-way THINGS-EEG2 protocols; encoder, preprocessing, checkpoint, and seed details still differ, so the table is contextual rather than a fully controlled ranking. Our subject-dependent row is the five-seed mean. Our LOSO row is a strict seed-42 re-evaluation and is marked accordingly.

The five seed-level subject-dependent macro means range from 85.60% to 87.35% Top-1 and from 98.20% to 98.55% Top-5, yielding $86.66 \pm 0.69\%$ and $98.38 \pm 0.14\%$. Our method is numerically strongest among the displayed intra-subject rows, but protocol differences preclude an unqualified state-of-the-art claim. In strict LOSO, the macro mean drops to $20.40 \pm 6.91\%$ Top-1 and $49.30 \pm 10.47\%$ Top-5 across held-out subjects. Figure 2 shows that all folds remain above chance but vary strongly, confirming that subject shift is still a central limitation.

Strict LOSO for Our method · 10 held-out subjects · seed 42

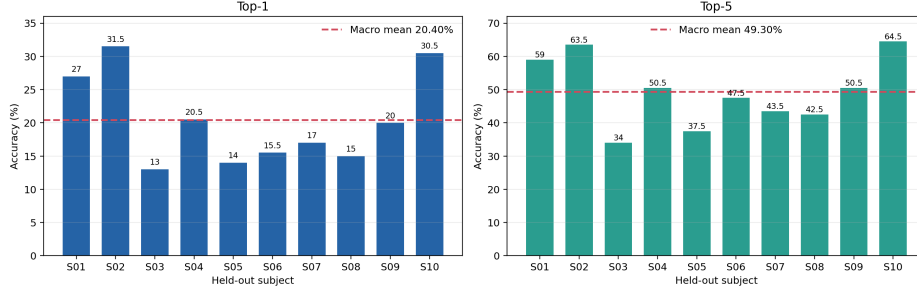


Figure 2: Strict leave-one-subject-out results at seed 42. Bars show independent-retrieval Top-1 and Top-5 for each held-out subject; dashed lines mark macro means. This is a single-seed generalization diagnostic rather than a five-seed population estimate.

Ablation Study: Effect of Image-Side Adaptation and TTUR

Single-subject pilot evidence for the proposed two-time-scale training strategy.

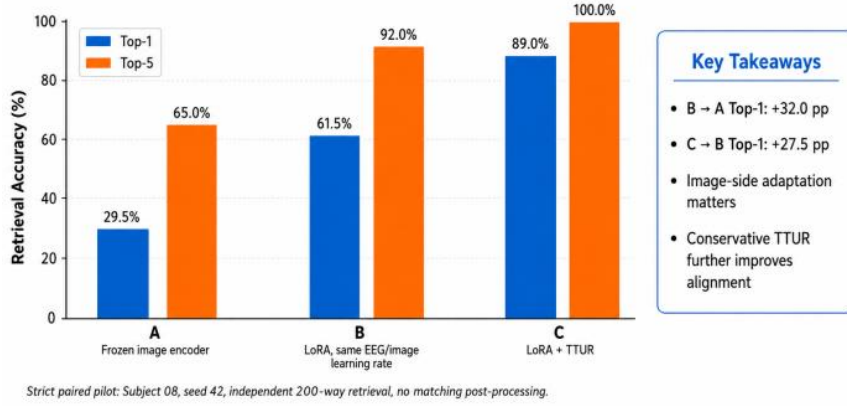


Figure 3: Strict paired subject-08/seed-42 TTUR ablation. All arms use the identical EEG encoder: A freezes the image encoder, B adds equal-rate image-side LoRA, and C uses slower image-side LoRA updates. Values are independent-retrieval accuracy (%).

5.2 Strict paired test of asymmetric co-adaptation

The A/B/C experiment directly addresses whether the improvement is merely caused by our EEG encoder. All three settings use subject 08, seed 42, the same EEG encoder architecture and initialization, identical samples, objective, batch order, and training budget. Only image-side optimization differs. Figure 3 shows that training the EEG encoder alone is insufficient: allowing image-side LoRA adaptation raises Top-1 by 32 points, and slowing the image update relative to the EEG update adds a further 27.5 points.

Relative to A, C gains +59.5 Top-1 and +35.0 Top-5 points. Because the EEG architecture is controlled, the result supports our proposed innovation: high performance depends on dual-side co-adaptation and its asymmetric update schedule, not only on a newly chosen EEG encoder. It does not, by itself, prove that 10:1 is universally optimal; the next experiment examines nearby settings.

5.3 Rank, learning-rate, and ratio study

The focused validation grid crosses LoRA rank $r \in \{32, 64\}$, EEG learning rate $\eta_e \in \{5 \times 10^{-4}, 10^{-3}\}$, and EEG-to-image learning-rate ratio $\rho \in \{5, 10, 20\}$, for 12 configurations on subject 08/seed 42. Table 2 reports validation scores. The highest validation Top-1 occurs at $(r, \eta_e, \rho) = (64, 10^{-3}, 20)$.

Table 2: Focused subject-08/seed-42 validation grid (%). Image LR is η_e/ρ . Bold denotes the Top-1-selected setting.

Rank	EEG LR	Ratio	Top-1	Top-5
32	5×10^{-4}	5	31.03	59.94
32	5×10^{-4}	10	31.52	60.24
32	5×10^{-4}	20	29.52	57.64
32	10^{-3}	5	29.76	57.58
32	10^{-3}	10	33.76	62.91
32	10^{-3}	20	32.42	61.82
64	5×10^{-4}	5	31.27	59.03
64	5×10^{-4}	10	32.06	60.91
64	5×10^{-4}	20	30.42	59.52
64	10^{-3}	5	27.39	55.88
64	10^{-3}	10	34.18	63.33
64	10^{-3}	20	34.48	63.03

Our method: focused hyperparameter confirmation · 10 subjects · seed 42

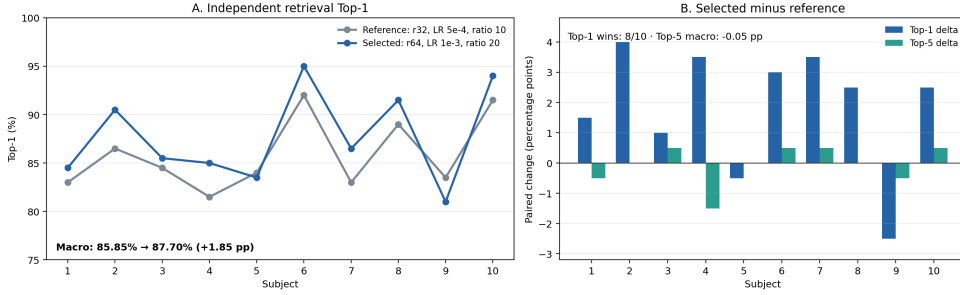


Figure 4: Formal ten-subject seed-42 confirmation of the validation-selected hyperparameter setting against the reference configuration. Top-1 improves by 1.85 points on average, while Top-5 is saturated and essentially unchanged.

We then compare the selected configuration with the reference ($32, 5 \times 10^{-4}, 10$) across three subject-08 validation seeds and finally all ten subjects at seed 42. The selected validation mean is 34.44%/63.74% versus 31.23%/60.97% for the reference. In the formal paired test, Figure 4 shows an average Top-1 increase from 85.85% to 87.70%; Top-5 changes from 98.35% to 98.30%. The selected setting wins Top-1 on eight of ten subjects, but it changes rank, absolute rate, and ratio jointly and uses one formal seed. The reference was protocol-locked before this focused study and is the only setting evaluated over all ten subjects and five seeds. We therefore do not retroactively replace the stronger-evidence headline with a post-hoc single-seed follow-up; the grid shows that asymmetric co-adaptation is robust to nearby choices, not that the original 10:1 ratio is globally optimal.

5.4 Fair comparison of matching algorithms

We evaluate NICE, ATM-S, and our method on subject 08/seed 42. Within each baseline, all five decoders receive the exact same EEG queries, image gallery, labels, checkpoint, and score matrix. Thus only the matching rule changes. Table 3 reports the standard 200×200 case. Independent Top-1/Top-5 remains the ordinary retrieval metric; the other columns are assignment accuracy.

The standard session rewards hard global consistency: Hungarian increases assignment accuracy by 13.5, 16.5, and 9.0 points over independent Top-1 for NICE, ATM-S, and our method. This does not mean the representation improved. It means the decoder used the external fact that every query and image should appear exactly once. Applying the same decoder to every baseline makes the comparison fair, but the result remains an offline structured-assignment result.

Table 3: Standard 200×200 fairness suite on subject 08/seed 42 (%). Structured columns are assignment accuracy, not retrieval Top-1.

Representation	Indep. T1	Indep. T5	Greedy	Hungarian	Stable	Sinkhorn [‡]
NICE	18.0	41.0	22.0	31.5	23.5	31.5
ATM-S	46.0	76.0	52.0	62.5	50.5	65.0
Our method	91.0	99.5	95.5	100.0	98.0	100.0

[‡]Raw-score Sinkhorn did not meet the preregistered 10^{-8} tolerance within 500 iterations in these three cells; values are diagnostic.

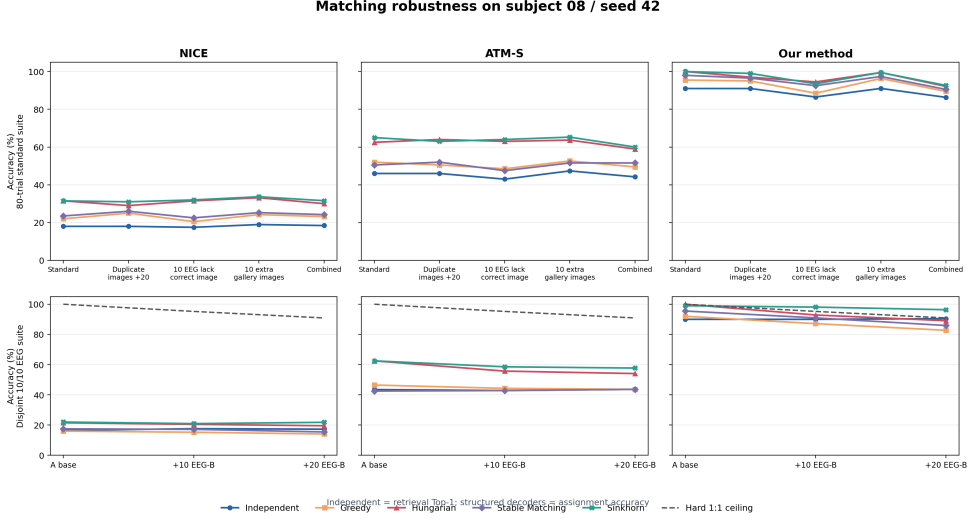


Figure 5: Matching robustness under missing, duplicate, and distractor items, including disjoint-trial duplicate EEG. Hard one-to-one assignment exposes unmatched queries when bijection is false.

The non-bijective suite in Figure 5 changes that conclusion. Removing only images creates unanswerable EEG queries; removing only EEG creates distractor gallery images; duplicate gallery entries create equivalent candidates; and combinations alter both matrix shape and answerability. The realistic disjoint-trial experiment is especially revealing. On the 200×200 20-trial-average base, independent/Hungarian/Sinkhorn achieve 90.0/100.0/99.0%. At 210×200 , they obtain 90.0/92.86/98.10%, with Hungarian leaving 10 queries unmatched. At 220×200 , they obtain 90.45/89.09/96.36%, with Hungarian leaving 20 queries unmatched. The hard bijection is therefore beneficial only when its premise is true; multiple legitimate EEG queries per image turn it into a constraint violation.

5.5 Sinkhorn convergence audit

Because Sinkhorn accuracy is meaningful only when its numerical behavior is known, we audit three score treatments over 90 cells each (three baselines by 30 scenarios): raw recorded scores; recorded-scale removal; and scale removal plus an epsilon schedule. Table 4 shows that scale removal greatly improves convergence but does not improve average assignment accuracy.

On the standard matrices, scale removal changes Sinkhorn assignment accuracy from 31.5% to 33.0% for NICE, from 65.0% to 57.0% for ATM-S, and leaves our method at 100.0%. The result rejects the simple hypothesis that better numerical convergence necessarily yields better matching. It also argues against spending the course-project budget on a learned matrix-transform network without a validation-defined objective and leakage-safe training split: convergence is a numerical property, whereas correct assignment depends on calibrated semantic structure.

Table 4: Sinkhorn audit over 90 baseline–scenario cells per setting.

Score treatment	Conv. 10^{-6}	Conv. 10^{-8}	Mean assignment
Raw recorded score	30/90	25/90	62.83%
Recorded scale removed	90/90	85/90	61.44%
Scale removed + epsilon schedule	90/90	85/90	61.44%

6 Discussion and Human-Centered Implications

What the system does and does not infer. Our evaluation decodes a stimulus from a known, closed 200-image gallery after repeated presentations. It is not open-world mind reading, does not reconstruct arbitrary thoughts, and does not establish semantic access beyond the experiment. Trial averaging uses substantial repeated evidence, so results should not be presented as single-glance decoding.

Privacy, consent, and agency. EEG is sensitive biometric and health-adjacent data. Deployment would require informed, revocable consent, purpose limitation, secure processing, restricted raw-signal retention, and clear uncertainty communication. Users should control calibration, deletion, and storage. This research system is not a medical device and should not support diagnosis, surveillance, employment screening, or covert inference.

Generalization and evaluation validity. The strict LOSO gap shows that cross-person transfer remains much weaker than subject-dependent calibration. Five seeds quantify optimization variability over the same stimuli, not new participants; literature rows also differ in encoders, preprocessing, and checkpoint rules. The subject-08 matching suite is diagnostic rather than population-level evidence, and batch matching can inflate apparent performance when a gallery is known to be a permutation. We therefore separate retrieval from assignment and test broken assumptions.

Technical limitations. Flattening discards explicit channel topology, and LoRA placement is not neurophysiologically targeted. The focused hyperparameter study is not a full factorial multi-seed search, while Sinkhorn row-wise decoding is a soft many-to-one diagnostic rather than a guaranteed discrete bijection. Future work should prioritize subject-invariant calibration, single-trial robustness, uncertainty, leakage-safe matcher calibration, and prospective studies of usefulness and trust.

7 Conclusion

Our asymmetric EEG–image co-adaptation reaches 86.66% Top-1 and 98.38% Top-5 over ten subjects and five seeds. With the EEG encoder fixed, the strict A/B/C ablation rises from 29.5% to 61.5% and 89.0%, isolating equal-rate and asymmetric image adaptation; focused tuning reaches 87.70%, while strict LOSO falls to 20.40%. Hungarian and Sinkhorn improve valid one-to-one assignments, not embeddings or standard Top-5, and weaken when correspondence fails.

Author contribution statement

Chi Kit Wong: overall brain–image co-adaptive framework and synchronized joint optimization with asymmetric learning rates. **Sida Zhang:** session-aware offline decoding as weighted bipartite matching and Hungarian/Sinkhorn analysis. **Zhiyi Chen:** lightweight residual-MLP EEG representation module and hyperparameter analysis. **Ziyi Guo:** CLIP/LoRA image adaptation and its conservative image-side learning rate. All members contributed equally to report writing, result interpretation, revision, and presentation preparation.

Generative-AI disclosure. OpenAI Codex assisted with code/log inspection, LaTeX/Markdown formatting, and language editing, including prompts to audit the page limit, reconcile saved CSV/JSON results, and repair equations. All outputs were reviewed and revised by the authors; drafts are retained in the project task history, and final claims remain the authors’ responsibility.

References

- [1] Alessandro T. Gifford, Kshitij Dwivedi, Gemma Roig, and Radoslaw M. Cichy. A large and rich EEG dataset for modeling human visual object recognition. *NeuroImage*, 264:119754, 2022. doi: 10.1016/j.neuroimage.2022.119754.
- [2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [3] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, volume 30, pages 6626–6637, 2017.
- [4] Yonghao Song, Bingchuan Liu, Xiang Li, Nanlin Shi, Yijun Wang, and Xiaorong Gao. Decoding natural images from EEG for object recognition. In *International Conference on Learning Representations*, 2024.
- [5] Dongyang Li, Chen Wei, Shiyang Li, Jiachen Zou, and Quanying Liu. Visual decoding and reconstruction via EEG embeddings with guided diffusion. In *Advances in Neural Information Processing Systems*, volume 37, pages 102822–102864, 2024.
- [6] Haitao Wu, Qing Li, Changqing Zhang, Zhen He, and Xiaomin Ying. Bridging the vision–brain gap with an uncertainty-aware blur prior. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2246–2257, 2025.
- [7] Wen Zhang, Shuai Wang, Yu Su, Xin Li, Changqing Zhang, and Shichao Zhong. NeuroBridge: Bio-inspired self-supervised EEG-to-image decoding via cognitive priors and bidirectional semantic alignment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(21):18028–18036, 2026. doi: 10.1609/aaai.v40i21.38863.
- [8] Yang Du, Siyuan Dai, Yonghao Song, Paul M. Thompson, Haoteng Tang, and Liang Zhan. Deep models, shallow alignment: Uncovering the granularity mismatch in neural decoding. *arXiv preprint arXiv:2601.21948*, 2026.
- [9] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.
- [10] Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
- [11] David Gale and Lloyd S. Shapley. College admissions and the stability of marriage. *The American Mathematical Monthly*, 69(1):9–15, 1962.
- [12] Richard Sinkhorn and Paul Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- [13] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.